

TEACHING STATEMENT

PAT LANK

0.1. **Philosophy.** As a first-generation student, I leaned heavily on mentorship and guidance of other people. This motivates me to feedback such experiences into my own teaching and mentorship practices. In particular, I align with Federico Ardila's Axioms:

- Mathematical talent is distributed equally among different groups, irrespective of geographic, demographic, and economic boundaries.
- Everyone can have joyful, meaningful, and empowering mathematical experiences.
- Mathematics is a powerful, malleable tool that can be shaped and used differently by communities to serve their needs.
- Every student deserves to be treated with dignity and respect.

In introductory courses, this includes building foundations by practice, addressing gaps in knowledge, and focusing on understanding rather than memorization. In advanced courses, this includes helping students see the path from coursework to independent work, and feeling the frontiers of knowledge. In mentorship, this includes helping students find their potential and providing the encouragement to pursue it.

0.2. **Experience.** I have been the instructor of record for fifteen course section. This ranges from college algebra to precalculus, business calculus, multivariable calculus, and discrete mathematics. In each course, I held sole responsibility for lectures, the syllabus, assignments, examinations, office hours, and grades.

At the start of a semester, I use a short in-person assessment to identify gaps in prerequisite knowledge. This helps identify reasonable learning goals of the course, decide which topics require review, and what supplementary material will be most useful.

During the semester, I use shorter assessments to identify problems before they appear on a major examination. In-person exercises show whether students can carry out the formal arguments. Short oral explanations and board work reveal whether they understand why those steps are valid. Group work activities help students discuss relevant material amongst each other.

I post notes, assignments, schedules, review material, and course announcements online and to my personal site. This allows students to easily find the relevant material. At the start of each lecture, I explicitly state the goal and outline the agenda.

My work with students was recognized by a teaching award from the Student Disability Resource Center at the University of South Carolina. I also served on a course-development committee for a college mathematics course and completed two years of pedagogical coursework. These experiences have made me more attentive curriculum development and refine my methods.

0.3. **Assessment and technology.** I design assessments to gauge a students reasoning, and not only whether the student has reached the expected conclusion. Depending on the course, I use a combination of examinations, homework, discussion, and board presentations. In proof-based courses, I would also give students opportunities to revise arguments after receiving feedback and to present them orally to me. Revision requires a student to identify what was missing and orally explain how the new argument resolves the problem.

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Artificial intelligence (AI) is changing how students produce work. In my courses, students could actively use AI to test examples, explore approaches, or search for gaps in an argument. They would nevertheless be required to explain the mathematics themselves and take responsibility for every step of a submitted solution.

As an instructor, I can benefit from AI tools. For instance, I could give students an incorrect or incomplete computer-generated output and ask them to repair it. In particular, I can seek methods which allow for computer-generated outputs that are catered individually to a student's weakness. I would continue seeking training in course design and assessment so that my teaching can respond to new tools without weakening standards.

0.4. Advanced teaching. My experience teaching introductory mathematics has prepared me to contribute broadly to the undergraduate curriculum. I am also prepared to teach advanced courses in algebraic geometry, commutative algebra, homological algebra, category theory, and derived categories. The experience I have with giving lectures in an advanced Summer School have helped me sharpen my communication of advanced concepts.

In this position, I would be interested in developing a two-semester algebraic geometry sequence. The first course would introduce affine and projective schemes, morphisms, fiber products, coherent sheaves, and basic dimension theory. It would begin with the commutative algebra needed to work with spectra and localization, while moving toward geometric examples.

The second course would cover sheaf cohomology, derived functors, base change, and related homological methods. Depending on the interests of the students, later topics could include divisors, duality, or deformation theory.

I would use *Algebraic Geometry I: Schemes* and *Algebraic Geometry II: Cohomology of Schemes* by Ulrich Görtz and Torsten Wedhorn as sources, together with the *Stacks Project*. This sequence would be to prepare students to work at the frontier of current research in algebraic geometry and commutative algebra.

I would also be happy to offer focused topics courses based on the interests of current and incoming students. Possible subjects include singularity theory, t -structures, generation and Rouquier dimension, Fourier–Mukai transforms, and homological methods in algebraic geometry.

0.5. Mentorship and training. I currently co-advise Elías Guisado Villalgordo, a PhD student at the Basque Center for Applied Mathematics, on relative Fourier–Mukai theory. This experience has reinforced the importance of dividing a large research problem into intermediate questions that allow a student to make visible progress. I want students to learn how to extract the main idea from a paper, test a proposed statement on examples, keep track of unresolved steps, and recognize when a failed approach has revealed useful information.

For the past five years, I have used *Zulip*, a team-chat platform organized by topic, to support collaborative research. Separate searchable threads allow a group to keep definitions, examples, proof attempts, references, and open questions distinct rather than distributing them across long email chains.

For a reading course or thesis project, the same structure can record weekly goals, questions arising between meetings, and the next steps agreed upon with the student. It supplements rather than replaces regular meetings and face-to-face mathematical discussion.

In this position, I would welcome the opportunity to supervise bachelor's, master's, and PhD theses. Where useful, I would develop co-supervised projects with researchers within my research network. This helps my students network with the mathematical community early on.

P. LANK, DIPARTIMENTO DI MATEMATICA “F. ENRIQUES”, UNIVERSITÀ DEGLI STUDI DI MILANO, VIA CESARE SARDINI 50, 20133 MILANO, ITALY

Email address: plankmathematics@gmail.com